



(ii) Determine the SI base units of η .

base units [2]

(c) The oil in (b) has a density of 920 kg m^{-3} and a viscosity of 4.7 in SI units.

The steel ball has a mass of $2.4 \times 10^{-3} \text{ kg}$ and a radius of $4.2 \times 10^{-3} \text{ m}$.

(i) Show that the upthrust force acting on the ball is $2.8 \times 10^{-3} \text{ N}$.

[1]

(ii) Determine the terminal speed v of the ball.

$v = \dots \text{ m s}^{-1}$ [3]

[Total: 12]

